Andrew Yuan

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Education

University of Massachusetts Amherst — Amherst, MA
Ph.D. Candidate in Computer Science — *Sept. 2023 – Present*
M.S. in Computer Science — *May 2026*

- Advised by Przemyslaw Grabowicz and David Jensen
- Research Interests: LLM Evaluation, Model Behavior, Generalization, Interpretable ML

Northeastern University — Boston, MA
B.S. in Computer Science, Minor in Mathematics — *Sept. 2019 – Jan 2023*

- GPA: 3.9/4.0

Research Experience

Doctoral Researcher — Sept. 2023 – Present
University of Massachusetts Amherst — *Amherst, MA*

- Doctoral research on language-model evaluation, interpretability, and generalization, with a focus on how experimental and benchmark design affect conclusions about model capabilities.

  Current:

  * Developing controlled evaluations of in-context learning to identify what information demonstrations provide and how demonstration selection affects model performance.
  * Developing automated methods for constructing executable Boolean queries under restrictive API constraints, comparing interpretable rule-learning and LLM-based approaches for targeted retrieval.

Research Assistant — Jan. 2022 – Apr. 2023
Northeastern University — *Boston, MA*

- Investigated how early transformer hidden states encode future token predictions by patching activations across model layers and measuring their effects on subsequent outputs.
- Developed and evaluated poisoning attacks targeting neural-network dropout; implemented vision models, designed experiments, and presented findings.

Publications

P. Pruthi, A. Yuan, A. D'Amour, and D. Jensen. "Evaluating Compositional Generalization in Transformers: The Role of Composition Equivalence and Module Coverage". *NeurIPS (Evaluations and Datasets Track)*, 2026

A. Yuan, A. Oprea, and C. Tan. "Dropout Attacks". *IEEE S&P*, 2024. Paper

K. Pal, J. Sun, A. Yuan, B. Wallace, and D. Bau. "Future Lens: Anticipating Subsequent Tokens from a Single Hidden State". *CoNLL*, 2023. Paper

Industry Experience

Cyber Security Graduate Intern — May. 2025 – Jul. 2025
MITRE — *Bedford, MA*

- Led the design and development of an end-to-end document ingestion pipeline combining computer vision, custom parsing, and retrieval-augmented LLMs to automate a previously manual workflow.
- Managed a small team of interns and drove the project through implementation and validation, resulting in a system designed to process thousands of documents with minimal oversight.

Full Stack Engineer Intern — Apr. 2020 – Dec. 2021
AbbVie — *Worcester, MA*

- Built scientific web applications for clonal-lineage analysis and single-cell RNA sequencing, developing interactive visualizations and working with scientists to translate research needs into application requirements.

Teaching

Teaching Assistant Sept. 2023 – Present
University of Massachusetts Amherst *Amherst, MA*

- CS220: Programming Methodology (Fall 2025)
- CS326: Web Programming (Summer, Fall 2024)
- CS589: Machine Learning (Spring 2024, Spring 2025)
- CS690F: Responsible Artificial Intelligence (Fall 2023)

Technical Skills

Languages: Python, TypeScript/JavaScript, Java, C, SQL
ML/AI: PyTorch, Hugging Face Transformers, scikit-learn, vLLM
Scientific Computing: NumPy, pandas, Matplotlib, Plotly
Systems & Tools: Slurm, Linux, Docker, Git